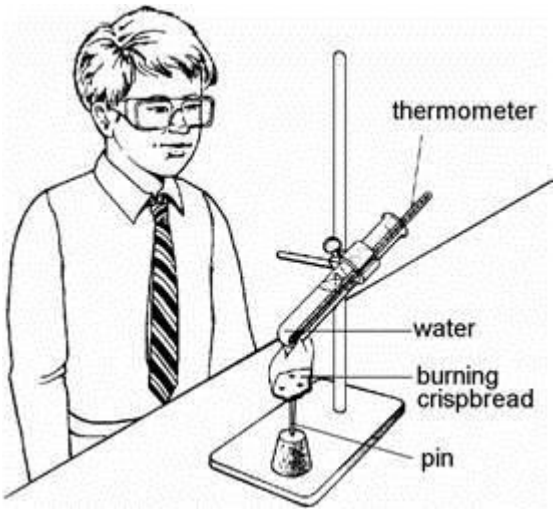
Q1.

Peter burns a piece of crispbread to find out how much energy is stored in it. Energy from the burning crispbread raises the temperature of the water in the test-tube.



(a) Describe one way Peter has arranged the apparatus so that he is working safely.

.....

.....

2 marks

(b) Peter wants to find out if potato crisps contain as much energy as crispbread. He does the experiment again using a piece of potato crisp.

Suggest **two** things he must do to make the experiment a fair test.

1. .....

2. .....

1 mark

The table shows some of the nutritional information from a packet of crispbread and a packet of potato crisps.

	energy in kJ	protein in g	carbohydrate in g	fat in g	fibre in g
100 g of crisp bread	1455	11.6	58.1	7.3	14.7
100 g of potato crisps	2072	5.8	57.9	28.7	4.3

(c) Peter burns 1.0 g of potato crisp instead of 1.0 g of crispbread in a similar experiment. What result will he get when he burns the potato crisp? Tick the correct box.

The change in the temperature of the water will be greater. ☐

The change in the temperature of the water will be the same. ☐

The change in the temperature of the water will be smaller. ☐

There will be no change in the temperature of the water. ☐

1 mark

(d) (i) Fibre contains energy. Explain why this energy can **not** be used by the human body.

.....

.....

1 mark

(ii) Use the table in part (b) to give **two** reasons for choosing crispbread rather than potato crisps as part of a balanced diet.

1. .....

.....

2. .....

.....

2 marks

(e) Crispbread does not contain vitamin C. Which of the foods in the list below is the best source of vitamin C?
Tick the correct box.

cheese	eggs	fish	oranges
☐	☐	☐	☐

1 mark
Maximum 8 marks

Q1.

(a) any **one** from

- the test-tube is pointing away from him
  *accept 'the test-tube is pointing away from the edge of the bench' do **not** accept 'the test-tube is at an angle'*
- he used a cork and a pin to hold the burning crispbread
  *accept 'he used a pin' **or** 'he is not holding the crispbread' **or** 'the cork is on the stand'*
- the test-tube is held in a clamp
  *accept 'he used a clamp' **or** 'the test-tube is held tightly'*
- it is away from the edge of the bench
- the apparatus is arranged over the base of the stand for stability
  accept 'the tube is over the base'
  *do **not** accept 'he is wearing goggles*

1 (L5)

(b) any **two** from

- use the same amount of water
- use the same mass **or** weight of crisp
  accept 'use the same mass of food'
  *do **not** accept 'use the same amount of food'*
- the crisp must be the same distance from the test tube as the crispbread was
- start with water at the same temperature
- shield both experiments from the draught
  *do **not** accept 'use the same apparatus'*
  ***or** 'heat for the same amount of time'*

2 (L4)

(c) The change in the temperature of the water will be greater. ✓
if more than one box is ticked, award no mark

1 (L6)

(d) (i) fibre is not digested
accept 'it is not absorbed'
***or** 'it does not get broken down'*
*do **not** accept 'it is insoluble'*

1 (L6)

(ii) any **two** from

- it contains less fat
  accept 'it is less fattening'
  or 'it contains less energy'
- it contains more fibre
- it contains more protein
  *do **not** accept 'more carbohydrate'*

2 (L6)

(e) oranges ✓

if more than one box is ticked, award no mark

1 (L5)

[8]